

Wildfire Susceptibility: a Random Forest Baseline

This notebook builds an end-to-end machine-learning pipeline that ranks wildfire susceptibility from static landscape predictors and yearly burned-area rasters.

Scientific context. Wildfire susceptibility mapping asks: given only terrain, vegetation, and accessibility, which parts of the landscape are most similar to places that have historically burned? The output is a *relative ranking*, not a calibrated annual ignition probability — weather, fuel moisture, and human ignition patterns are not modelled here.

Pipeline overview:

#	Section	Purpose
1	Setup	Environment, constants, paths
2	Data loading	Raster ingestion and analysis mask
3	Burned-area target	Fire rasters and temporal split
4	Feature engineering	Aspect encoding, vegetation one-hot
5	Training set construction	Pseudo-absences and train/val split
6	Exploratory class comparison	Burned vs unburned contrasts
7	Model tuning	Hyperparameter grid search
8	Model fitting	Final pre-2016 model
9	Hold-out evaluation	ROC-AUC, recall@top%, score distributions
10	Threshold decisions	Percentile-based operational thresholds
11	Model explanation	Permutation importance (MDA)
12	Susceptibility map	Spatial prediction and overlay

Key design choices:

- The row unit is one valid 100 m pixel.
- Positive label: pixel burned at least once in the period.
- Training uses balanced pseudo-absences; hold-out keeps real imbalance.
- The model output is a relative susceptibility score.
- Main caveats: nearby pixels are spatially autocorrelated, predictors are assumed time-invariant.

1. Setup

This cell loads the lesson setup script. In Colab it downloads the script from GitHub; the script then installs dependencies and clones the data.

Features as they appear in the raw rasters.

```
In [1]: from pathlib import Path
from urllib.request import urlretrieve

setup_path = Path("lesson_setup.py")
if not setup_path.exists():
    urlretrieve(
        "https://raw.githubusercontent.com/mirkodandrea/wildfire-susceptibil
        setup_path,
    )

import lesson_setup  # installs packages and clones the data repo when runni

# --- core scientific Python stack ---
import numpy as np          # numerical arrays and math
import pandas as pd         # tabular data – DataFrames and Series
import matplotlib.pyplot as plt  # plotting

import rasterio  # read and write GeoTIFF and other geospatial raster format

# --- notebook display helpers ---
from IPython.display import display, Markdown  # render formatted output in

# --- matplotlib colour utilities (used in the vegetation raster map) ---
from matplotlib.colors import BoundaryNorm, ListedColormap  # colour scales
from matplotlib.patches import Patch  # custom legend

plt.style.use("seaborn-v0_8-whitegrid")
pd.set_option("display.max_columns", 100)
pd.set_option("display.width", 140)

RANDOM_STATE = 42
HOLDOUT_START_YEAR = 2016

DATA_DIR = Path("data")
FIRE_RASTER_DIR = DATA_DIR / "fires"

RASTER_PATHS = {
    "elevation": DATA_DIR / "elevation.tif",
    "slope": DATA_DIR / "slope.tif",
    "aspect": DATA_DIR / "aspect.tif",
    "vegetation": DATA_DIR / "vegetation.tif",
    "urban_distance": DATA_DIR / "urban_distance.tif",
    "roads_distance": DATA_DIR / "roads_distance.tif",
}

# Features as they appear in the raw rasters.
RASTER_FEATURES = [
    "elevation",
```

```

        "slope",
        "aspect",
        "urban_distance",
        "roads_distance",
        "vegetation",
    ]

    # Features as they enter the model (after engineering).
    FEATURES = [
        "elevation",
        "slope",
        "aspect_eastness",
        "aspect_northness",
        "urban_distance",
        "roads_distance",
        "vegetation",
    ]

def continuous_features(feature_list):
    """Return only the continuous (non-categorical) features.

    Vegetation is categorical and handled separately via one-hot encoding.
    """
    return [f for f in feature_list if f != "vegetation"]

VEGETATION_NAMES = {
    # Mixed broadleaf forests
    3112: "Warm-climate mixed forest",
    3113: "Moist-climate mixed forest",
    313: "Mixed forest",
    # Dominant tree-species forests
    3114: "Beech forest",
    3115: "Chestnut forest",
    3116: "Chestnut orchard",
    # Evergreen and conifer forests
    3111: "Evergreen dry forest",
    312: "Coniferous forest",
    # Water-associated vegetation
    3117: "Riparian vegetation",
    # Shrub and transitional vegetation
    322: "Shrubland and scrub",
    323: "Mediterranean evergreen scrub",
    324: "Woodland-shrub transition",
}

VEGETATION_CODES = list(VEGETATION_NAMES)
VEGETATION_FEATURE_COLUMNS = [f"vegetation_{code}" for code in VEGETATION_CODES]

# Colors for vegetation classes, chosen to be visually distinct
VEGETATION_COLORS = {
    3112: "#5a9e61",
    3113: "#1b7837",
    313: "#00441b",
    3114: "#74c476",

```

```

3115: "#238b45",
3116: "#8c6d31",
3111: "#2b8c4b",
312: "#006d2c",
3117: "#2b8cbe",
322: "#d8b365",
323: "#b35806",
324: "#a6d96a",
}
VEGETATION_CMAP = ListedColormap([VEGETATION_COLORS[code] for code in VEGETATION_CODES])
VEGETATION_NORM = BoundaryNorm(
    np.arange(-0.5, len(VEGETATION_CODES) + 0.5),
    VEGETATION_CMAP.N,
)

def clean_axes(ax):
    """Remove top and right spines for a cleaner plot."""
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)

```

2. Data loading and analysis mask

Learning objective: understand how raster grids become arrays, and why a consistent validity mask is essential.

All rasters share the same 100 m grid. The `vegetation` raster defines the reference grid shape and coordinate transform. Each raster is read as a 2-D NumPy array.

The **analysis mask** restricts all subsequent operations to pixels where every predictor is available and meaningful. Pixels with missing vegetation (or code 0, which indicates non-vegetated/urban areas) are excluded first; then any pixel where a continuous predictor is missing is also excluded. This ensures a consistent footprint for sampling, evaluation, and mapping.

```

In [2]: with rasterio.open(RASTER_PATHS["vegetation"]) as src:
        TEMPLATE_TRANSFORM = src.transform
        TEMPLATE_SHAPE = (src.height, src.width)

        extent = (
            TEMPLATE_TRANSFORM.c,
            TEMPLATE_TRANSFORM.c + TEMPLATE_TRANSFORM.a * TEMPLATE_SHAPE[1],
            TEMPLATE_TRANSFORM.f + TEMPLATE_TRANSFORM.e * TEMPLATE_SHAPE[0],
            TEMPLATE_TRANSFORM.f,
        )

        raster_arrays = {}
        for name, path in RASTER_PATHS.items():
            with rasterio.open(path) as src:
                raster_arrays[name] = src.read(1, masked=True).astype("float32").fill(

```

```
In [3]: vegetation = raster_arrays["vegetation"]
analysis_mask = np.isfinite(vegetation) & (vegetation != 0)

valid_mask = analysis_mask.copy()
for feature in continuous_features(RASTER_FEATURES):
    raster_arrays[feature] = np.where(analysis_mask, raster_arrays[feature],
    valid_mask &= np.isfinite(raster_arrays[feature])

valid_pixel_count = int(valid_mask.sum())
display(Markdown("### Valid analysis pixels"))
display(valid_pixel_count)
```

Valid analysis pixels

398697

Predictor maps

Mapping each predictor is a quick sanity check: look for spatial structure, missing-data holes, etc.

```
In [4]: fig, axes = plt.subplots(
    len(continuous_features(RASTER_FEATURES)),
    1,
    figsize=(8, 3 * len(continuous_features(RASTER_FEATURES))),
    constrained_layout=True,
)
axes = np.atleast_1d(axes)

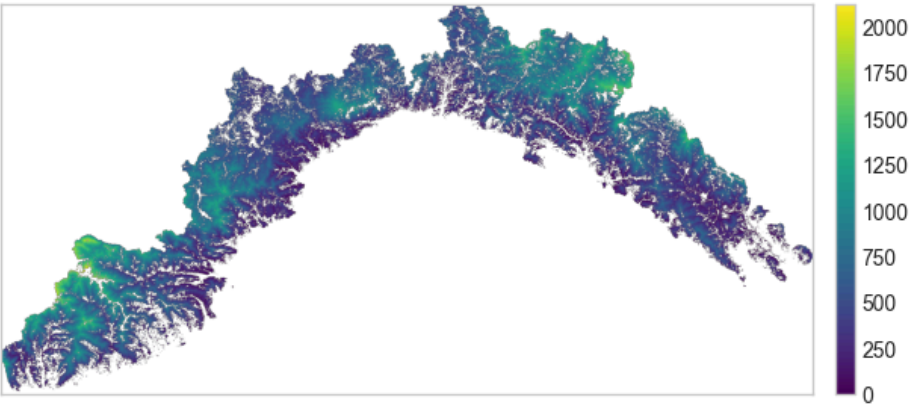
for ax, feature in zip(axes, continuous_features(RASTER_FEATURES)):
    feature_map = np.where(analysis_mask, raster_arrays[feature], np.nan)
    image = ax.imshow(feature_map, extent=extent, origin="upper", cmap="viridis")
    fig.colorbar(image, ax=ax, fraction=0.04, pad=0.02)
    ax.set_title(feature.replace("_", " "), loc="left", fontweight="bold")
    ax.set_xticks([])
    ax.set_yticks([])

fig.suptitle("Continuous predictor rasters", x=0.01, ha="left", fontweight="bold")
```

Out[4]: Text(0.01, 0.98, 'Continuous predictor rasters')

Continuous predictor rasters

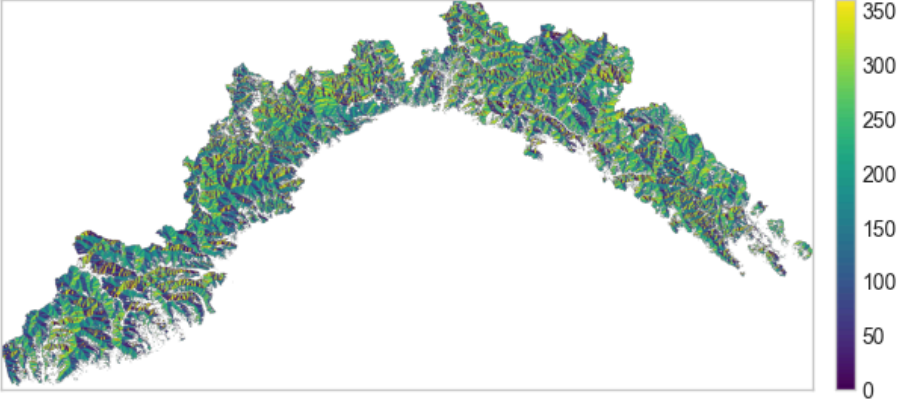
elevation



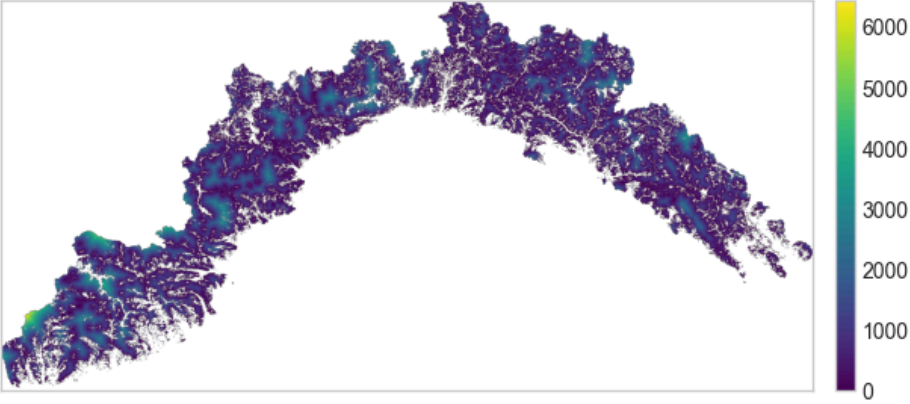
slope



aspect

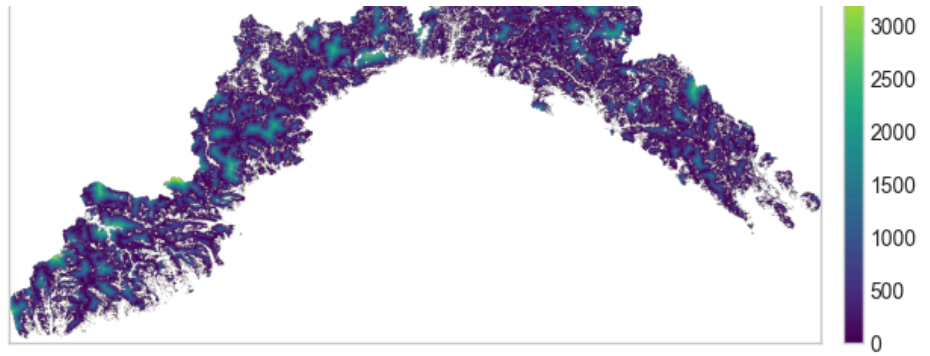


urban distance



roads distance





```
In [5]: fig, ax = plt.subplots(figsize=(11, 4.5), constrained_layout=True)
feature_map = np.where(analysis_mask, raster_arrays["vegetation"], np.nan)
code_to_index = {code: index for index, code in enumerate(VEGETATION_CODES)}
vegetation_map = np.full(feature_map.shape, np.nan, dtype=float)
for code, index in code_to_index.items():
    vegetation_map[feature_map == code] = index

ax.imshow(
    vegetation_map,
    extent=extent,
    origin="upper",
    cmap=VEGETATION_CMAP,
    norm=VEGETATION_NORM,
)
legend_handles = [
    Patch(facecolor=VEGETATION_COLORS[code], edgecolor="none", label=VEGETAT
    for code in VEGETATION_CODES
    if np.any(feature_map == code)
]
ax.legend(
    handles=legend_handles,
    title="Vegetation class",
    loc="center left",
    bbox_to_anchor=(1.01, 0.5),
    frameon=False,
)
ax.set_title("Vegetation", loc="left", fontweight="bold")
ax.set_xticks([])
ax.set_yticks([])
```

Out[5]: []

Vegetation



3. Burned-area target and temporal split

Learning objective: understand how yearly fire rasters become supervised labels, and why the temporal split matters for honest evaluation.

Each yearly fire raster is a binary grid: `1` = pixel burned that year, `0` = not burned. We combine years with a logical OR to get period masks.

The **temporal split** separates training data (fires before 2016) from hold-out evaluation (fires 2016–2022). This simulates an operational scenario: can a model trained on past fire patterns rank future fire locations?

⚠ **Assumption:** predictor rasters (elevation, slope, vegetation, distances) are treated as time-invariant. In reality, vegetation and urban extent change over 25 years. This approximation is acceptable for a baseline but should be noted when interpreting results.

```
In [6]: fire_years = list(range(1997, 2023))

fire_masks = {}
for year in fire_years:
    path = FIRE_RASTER_DIR / f"fire_{year}.tiff"
    with rasterio.open(path) as src:
        fire_mask = src.read(1).astype(bool)
        fire_masks[year] = fire_mask & valid_mask

pixel_area_km2 = abs(TEMPLATE_TRANSFORM.a * TEMPLATE_TRANSFORM.e) / 1_000_000
fire_summary_df = pd.DataFrame(
    [
        {
            "fire_year": year,
            "burned_pixels": int(mask.sum()),
            "burned_area_km2": round(int(mask.sum()) * pixel_area_km2, 2),
        }
        for year, mask in fire_masks.items()
    ]
)
```

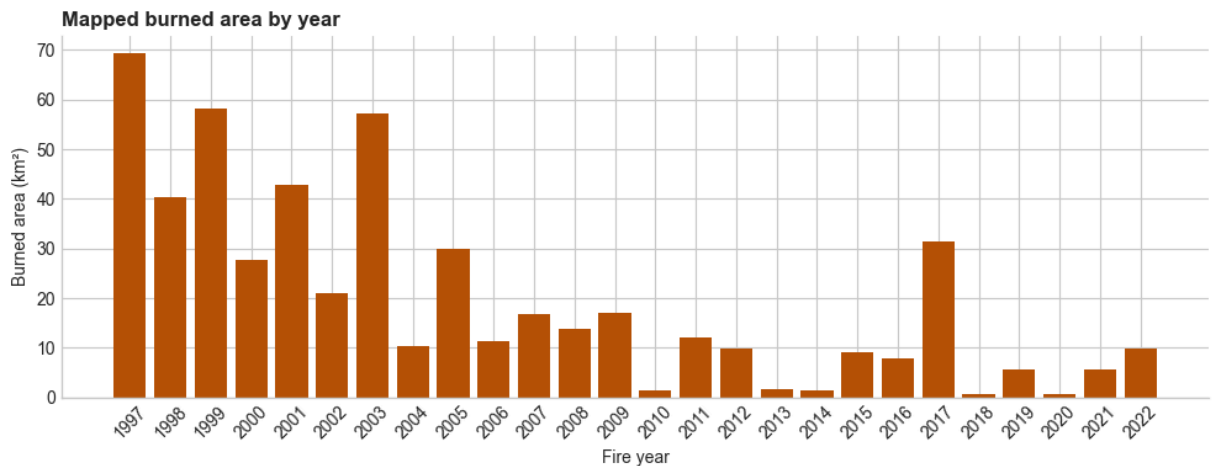


```

)

fig, ax = plt.subplots(figsize=(10, 4))
ax.bar(
    fire_summary_df["fire_year"].astype(str),
    fire_summary_df["burned_area_km2"],
    color="#b45309",
)
ax.set_title("Mapped burned area by year", loc="left", fontweight="bold")
ax.set_xlabel("Fire year")
ax.set_ylabel("Burned area (km²)")
ax.tick_params(axis="x", rotation=45)
clean_axes(ax)
fig.tight_layout()

```



Define the temporal split

```

In [7]: train_years = [year for year in fire_years if year < HOLDOUT_START_YEAR]
test_years = [year for year in fire_years if year >= HOLDOUT_START_YEAR]

train_burned_mask = np.logical_or.reduce([fire_masks[year] for year in train_years])
test_burned_mask = np.logical_or.reduce([fire_masks[year] for year in test_years])

temporal_split_df = pd.DataFrame(
    [
        {
            "role": "training pool",
            "years": f"{min(train_years)}--{max(train_years)}",
            "n_years": len(train_years),
            "burned_pixels": int(train_burned_mask.sum()),
            "burned_share_%": round(float(train_burned_mask.sum() / valid_pixels), 2),
        },
        {
            "role": "hold-out test",
            "years": f"{min(test_years)}--{max(test_years)}",
            "n_years": len(test_years),
            "burned_pixels": int(test_burned_mask.sum()),
            "burned_share_%": round(float(test_burned_mask.sum() / valid_pixels), 2),
        },
    ]
)

```

```
)
display(Markdown("### Temporal split"))
display(temporal_split_df)
```

Temporal split

	role	years	n_years	burned_pixels	burned_share_%
0	training pool	1997–2015	19	35912	9.01
1	hold-out test	2016–2022	7	5806	1.46

```
In [8]: fig, axes = plt.subplots(2, 1, figsize=(8, 8), constrained_layout=True)

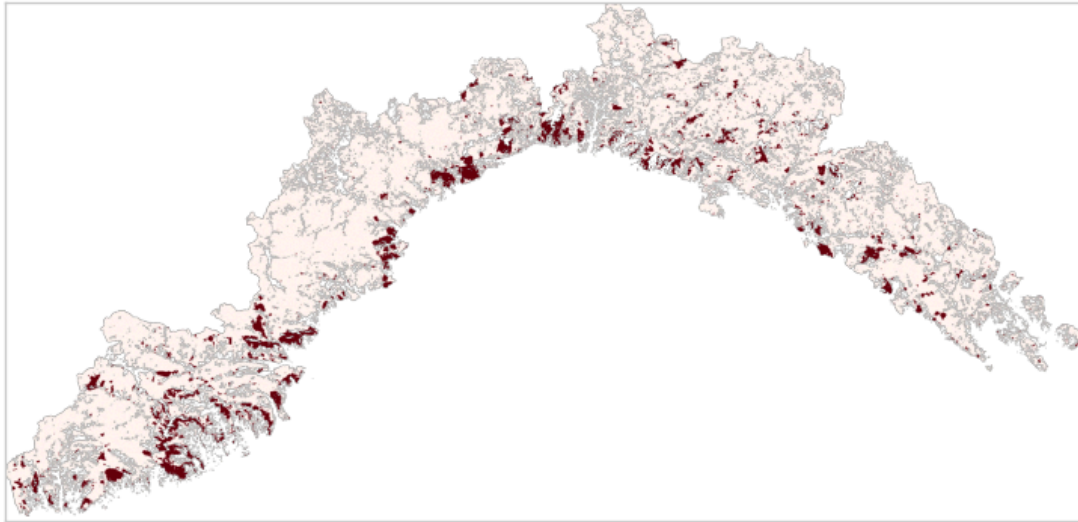
for ax, period_name, label_mask in [
    (axes[0], f"{min(train_years)}–{max(train_years)} (training)", train_burned),
    (axes[1], f"{min(test_years)}–{max(test_years)} (hold-out)", test_burned)
]:
    label_map = np.where(valid_mask, label_mask.astype(float), np.nan)
    image = ax.imshow(label_map, extent=extent, origin="upper", cmap="Reds",
    ax.set_title(period_name, loc="left", fontweight="bold")
    ax.set_xticks([])
    ax.set_yticks([])

fig.colorbar(image, ax=axes, fraction=0.025, pad=0.02, label="Burned (1) / L")
fig.suptitle("Burned labels by period", x=0.01, ha="left", fontweight="bold")
```

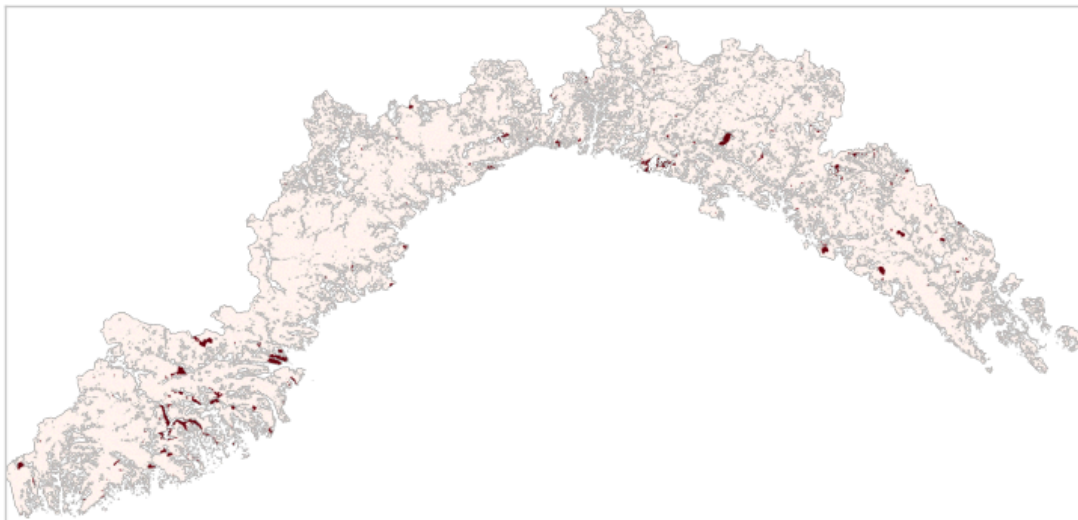
```
Out[8]: Text(0.01, 0.98, 'Burned labels by period')
```

Burned labels by period

1997–2015 (training)



2016–2022 (hold-out)



4. Feature engineering

Learning objective: understand why raw raster values cannot always be used directly, and how to transform circular and categorical predictors.

Two predictors need transformation before modelling:

Aspect (circular variable)

Aspect is measured in degrees (0–360). The problem: 1° and 359° are nearly the same direction (north), yet a model would see them as 358 units apart. The standard fix decomposes the angle into two orthogonal components:

- $\text{aspect_eastness} = \sin(\text{aspect}^\circ)$ → ranges from -1 (west) to +1 (east)
- $\text{aspect_northness} = \cos(\text{aspect}^\circ)$ → ranges from -1 (south) to +1 (north)

These two features together preserve direction information without any artificial discontinuity.

Vegetation (unordered categorical)

Vegetation codes (e.g., 3111, 322) have no meaningful numeric ordering. Treating them as a continuous variable would imply that code 3111 is "between" codes 312 and 3112, which is ecologically meaningless.

One-hot encoding creates one binary column per vegetation class. Each pixel gets a `1` in the column matching its class and `0` elsewhere. This lets the model learn class-specific effects without assuming any ordering.

For tree-based models, one-hot encoding is not strictly required (ordinal encoding also works), but it makes feature importance directly interpretable per vegetation class, and it is the standard approach in scikit-learn.

```
In [9]: def add_aspect_features(frame):
        """Replace raw aspect degrees with eastness and northness components."""
        engineered = frame.copy()
        aspect_radians = np.deg2rad(engineered["aspect"])
        engineered["aspect_eastness"] = np.sin(aspect_radians)
        engineered["aspect_northness"] = np.cos(aspect_radians)
        return engineered

def one_hot_encode_vegetation(frame):
    """Return one binary column per vegetation class from the fixed class list"""
    vegetation_codes = frame["vegetation"].astype(int)
    vegetation_dummies = pd.get_dummies(
        vegetation_codes,
        prefix="vegetation",
        dtype=int,
    )
    return vegetation_dummies.reindex(
        columns=VEGETATION_FEATURE_COLUMNS,
        fill_value=0,
    ).reset_index(drop=True)

def make_features(frame, feature_columns=None):
    """Build the numeric model matrix used by the Random Forest.

    Steps:
    1. Compute aspect_eastness and aspect_northness (if not already present)
    2. Select the continuous engineered features.
    3. One-hot encode vegetation.
    4. If feature_columns is provided, reindex to match training columns exactly
    """
    has_aspect_components = {"aspect_eastness", "aspect_northness"}.issubset
```

```

needs_aspect_engineering = "aspect" in frame.columns and not has_aspect_
engineered_frame = add_aspect_features(frame) if needs_aspect_engineerir

numeric_features = engineered_frame[continuous_features(FEATURES)].reset
vegetation_features = one_hot_encode_vegetation(engineered_frame)

model_features = pd.concat([numeric_features, vegetation_features], axis

if feature_columns is not None:
    model_features = model_features.reindex(columns=feature_columns, fil

return model_features

```

Build the shared feature table

One row per valid pixel, with coordinates and all engineered predictors. This table is built once and then joined with period-specific labels.

```

In [10]: def pixel_frame(
    rows,
    cols,
    target,
    sample_type,
    template_transform,
    raster_arrays,
    features,
):
    """Convert raster row/column indices into a tabular pixel dataset."""
    rows = np.asarray(rows, dtype=int)
    cols = np.asarray(cols, dtype=int)

    frame = pd.DataFrame(
        {
            "target": int(target),
            "sample_type": sample_type,
            "row": rows,
            "col": cols,
            "x": template_transform.c + (cols + 0.5) * template_transform.a,
            "y": template_transform.f + (rows + 0.5) * template_transform.e,
        }
    )

    for feature in features:
        values = raster_arrays[feature][rows, cols]
        frame[feature] = values.astype(int) if feature == "vegetation" else

    return frame

rows, cols = np.where(valid_mask)
feature_table = pixel_frame(
    rows,
    cols,
    target=0,
    sample_type="valid_pixel",

```

```

    template_transform=TEMPLATE_TRANSFORM,
    raster_arrays=raster_arrays,
    features=RASTER_FEATURES,
)
feature_table = add_aspect_features(feature_table)
feature_table = feature_table.drop(columns=["target", "sample_type"])

display(Markdown("### Feature table (first rows)"))
display(feature_table.head())

```

Feature table (first rows)

	row	col		x	y	elevation	slope	aspect	urban_di
0	1	1158	1.494904e+06	4.946934e+06		563.0	20.843185	174.158386	900.0
1	2	1156	1.494704e+06	4.946834e+06		527.0	23.599121	168.111343	824.0
2	2	1157	1.494804e+06	4.946834e+06		525.0	26.699856	175.295486	806.0
3	2	1158	1.494904e+06	4.946834e+06		519.0	24.512413	183.772278	800.0
4	2	1159	1.495004e+06	4.946834e+06		532.0	20.421495	194.782715	800.0

5. Training set construction

Learning objective: understand pseudo-absence sampling and why class balance matters for training.

Why pseudo-absences?

Fire is rare: typically only 1–5% of the landscape burns in any period. If we trained on all pixels, the model could achieve 95%+ accuracy by predicting "unburned" everywhere — useless for susceptibility ranking.

Pseudo-absence sampling selects a subset of unburned pixels as the negative class. We use a 1:1 ratio (equal burned and unburned samples) because:

1. It gives both classes equal weight during fitting, forcing the model to learn discriminative features rather than base-rate shortcuts.
2. It is the simplest balanced design and a standard baseline in species distribution modelling (which shares this structure).
3. The model's predicted scores no longer reflect landscape prevalence — but we explicitly want a *ranking*, not a probability estimate.

Alternative ratios (1:2, 1:5, 1:10) shift the model toward higher specificity at the cost of sensitivity. For a first baseline, 1:1 is the most transparent choice.

```

In [11]: def add_period_labels(features, burned_mask, period_name):
          """Join the feature table with a period-specific burned mask."""

```

```

period_df = features.copy()
period_df["target"] = burned_mask[period_df["row"], period_df["col"]].as
period_df["sample_type"] = np.where(period_df["target"] == 1, "burned",
period_df["period"] = period_name
period_df["target_name"] = period_df["target"].map({0: "unburned", 1: "b
return period_df

# Label all valid pixels for training period (needed to sample from).
train_period_df = add_period_labels(
    feature_table,
    train_burned_mask,
    f"{min(train_years)}--{max(train_years)}",
)

# Balanced training pool: all burned + equal random sample of unburned.
burned_train_df = train_period_df.loc[train_period_df["target"] == 1]
unburned_train_df = train_period_df.loc[train_period_df["target"] == 0].samp
    n=len(burned_train_df),
    random_state=RANDOM_STATE,
)

# Shuffle the combined training pool to mix burned and unburned samples.
train_pool_df = pd.concat([burned_train_df, unburned_train_df], ignore_index
    .sample(frac=1, random_state=RANDOM_STATE).reset_index(drop=True)

display(Markdown("### Training pool sample (balanced 1:1)"))
display(train_pool_df.head())

```

Training pool sample (balanced 1:1)

	row	col	x	y	elevation	slope	aspect	urban_di
0	317	768	1.455904e+06	4.915334e+06	581.0	19.239143	82.798622	1341.0
1	206	1323	1.511404e+06	4.926434e+06	733.0	18.362600	273.670532	0.0
2	649	541	1.433204e+06	4.882134e+06	184.0	19.080738	184.144623	806.0
3	926	99	1.389004e+06	4.854434e+06	221.0	19.826370	162.444519	565.0
4	251	1085	1.487604e+06	4.921934e+06	267.0	33.381763	50.542347	360.0

6. Exploratory class comparison

Learning objective: check whether burned and unburned pixels have distinguishable predictor distributions *before* fitting a model.

These summaries use the balanced pre-2016 training pool. They describe the modelling contrast (burned vs sampled pseudo-absence), not the true landscape prevalence of

each predictor value.

```
In [12]: class_summary_df = (
    train_pool_df.groupby("target_name", observed=True)[continuous_features(
        ).median()
        ].T.rename_axis("feature")
        .reset_index()
    )
    display(Markdown("### Class-wise median (continuous predictors)"))
    display(class_summary_df.style.format(precision=2))
```

Class-wise median (continuous predictors)

target_name	feature	burned	unburned
0	elevation	467.00	565.00
1	slope	21.34	18.80
2	aspect_eastness	0.16	0.00
3	aspect_northness	-0.56	0.00
4	urban_distance	583.10	583.10
5	roads_distance	316.23	300.00

```
In [13]: vegetation_summary_df = (
    pd.crosstab(
        train_pool_df["vegetation"].map(VEGETATION_NAMES).fillna(
            train_pool_df["vegetation"].astype(str)
        ),
        train_pool_df["target_name"],
        normalize="columns",
    )
    .mul(100)
    .round(1)
    .sort_values("burned", ascending=False)
    .head(8)
)
    display(Markdown("### Top vegetation classes (% within each label)"))
    display(vegetation_summary_df)
```

Top vegetation classes (% within each label)

target_name	burned	unburned
vegetation		
Warm-climate mixed forest	21.0	15.9
Mediterranean evergreen scrub	18.7	1.7
Shrubland and scrub	13.3	2.8
Mixed forest	11.3	13.7
Chestnut forest	10.7	26.7
Coniferous forest	10.1	8.4
Woodland-shrub transition	6.3	5.6
Moist-climate mixed forest	4.5	12.4

Distribution plots

Blue = sampled unburned (pseudo-absence). Red = burned.

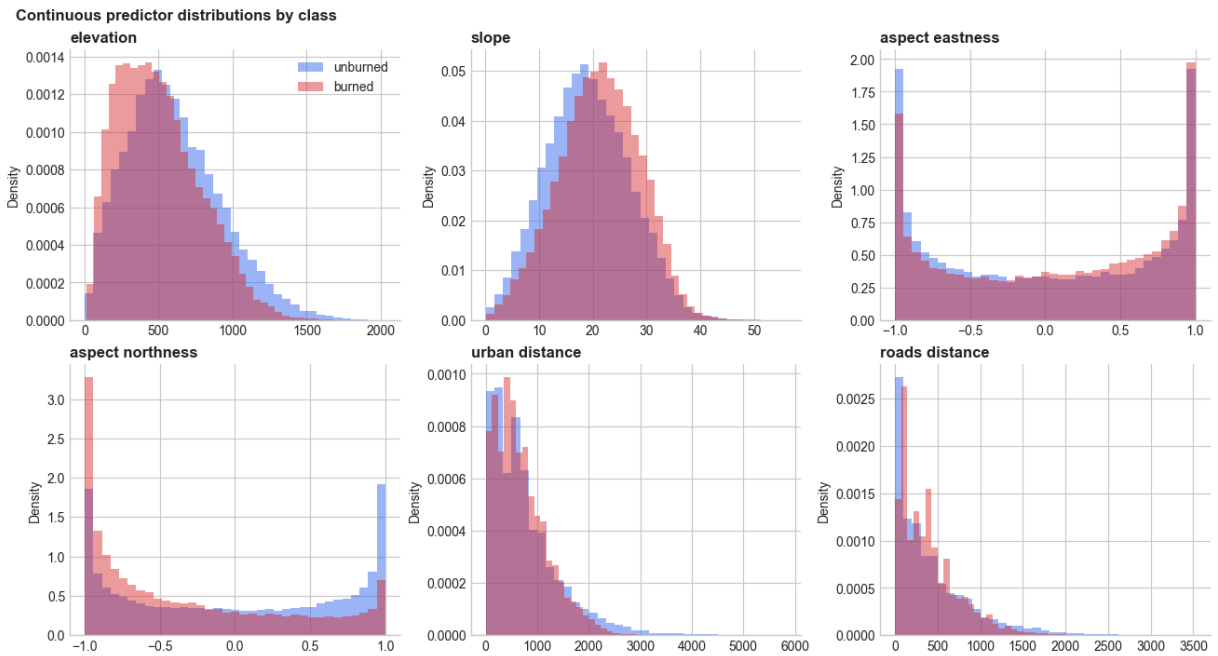
Strong separation suggests a useful predictor; heavy overlap means the model needs feature combinations rather than single thresholds.

```
In [14]: fig, axes = plt.subplots(2, 3, figsize=(13, 7), constrained_layout=True)
axes = axes.ravel()

for ax, feature in zip(axes, continuous_features(FEATURES)):
    for target, label, color in [(0, "unburned", "#2563eb"), (1, "burned", "#d62728")]:
        ax.hist(
            train_pool_df.loc[train_pool_df["target"] == target, feature],
            bins=35,
            density=True,
            alpha=0.45,
            color=color,
            label=label,
        )
    ax.set_title(feature.replace("_", " "), loc="left", fontweight="bold")
    ax.set_ylabel("Density")
    clean_axes(ax)

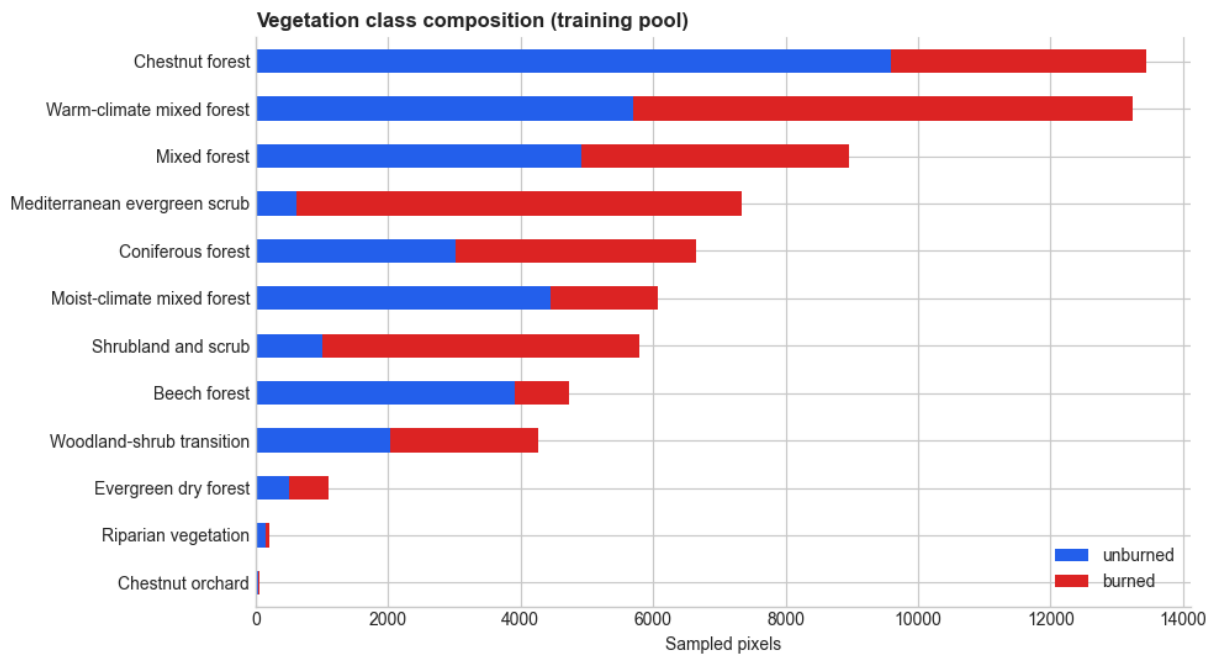
axes[0].legend(frameon=False)
fig.suptitle("Continuous predictor distributions by class", x=0.01, ha="left")
```

```
Out[14]: Text(0.01, 0.98, 'Continuous predictor distributions by class')
```



```
In [15]: vegetation_plot_df = (
    pd.crosstab(
        train_pool_df["vegetation"].map(VEGETATION_NAMES).fillna(
            train_pool_df["vegetation"].astype(str)
        ),
        train_pool_df["target_name"],
    )
    .reindex(columns=["unburned", "burned"], fill_value=0)
    .assign(total=lambda frame: frame.sum(axis=1))
    .sort_values("total", ascending=True)
)
vegetation_count_plot_df = vegetation_plot_df[["unburned", "burned"]]

fig, ax = plt.subplots(figsize=(10, 5.5))
vegetation_count_plot_df.plot.barh(stacked=True, ax=ax, color=["#2563eb", "#f8719d"])
ax.set_title("Vegetation class composition (training pool)", loc="left", fontweight="bold")
ax.set_xlabel("Sampled pixels")
ax.set_ylabel("")
clean_axes(ax)
ax.legend(title="", frameon=False)
fig.tight_layout()
```



Pre-2016 burned pixels tend to be lower in elevation (467 m vs 565 m), steeper (21.3° vs 18.8°), and more south-facing (aspect_northness -0.56 vs 0.00) than the sampled unburned pixels. Mediterranean evergreen scrub (18.7% vs 1.7%) and shrubland (13.3% vs 2.8%) are strongly over-represented among burned samples, while chestnut forests (10.7% vs 26.7%) and moist-climate forests (4.5% vs 12.4%) are under-represented. Warm-climate mixed forests show moderate enrichment in burned pixels (21.0% vs 15.9%), suggesting vegetation type is a key discriminator for susceptibility ranking.

Hold-out set: no balancing

With the training pool explored above, we can now finalise the dataset structure. The 2016–2022 hold-out set keeps *every* valid pixel. This preserves the real rare-event imbalance so that evaluation metrics reflect operational performance, not artificial balance.

```
In [16]: test_df = add_period_labels(
    feature_table,
    test_burned_mask,
    f"{min(test_years)}--{max(test_years)}",
)
```

Train / validation split

The validation split is internal to the balanced pre-2016 training pool. Stratification preserves the 50/50 ratio in both splits. This validation set is used only for hyperparameter tuning; the 2016–2022 hold-out remains untouched until final evaluation.

⚠ **Spatial autocorrelation caveat:** this random split does not respect spatial proximity. Nearby pixels share terrain, vegetation, and fire history, so validation AUC from this split is likely *optimistic* compared to truly independent spatial evaluation. The temporal hold-out (2016–2022) provides a harder test. This is acceptable as a first baseline for a lesson, a better approach would be to use spatial blocks or a spatial cross-validation strategy.

```
In [17]: from sklearn.model_selection import train_test_split # randomly split rows

train_model_df, validation_df = train_test_split(
    train_pool_df,
    test_size=0.2,
    random_state=RANDOM_STATE,
    stratify=train_pool_df["target"],
)
```

```
In [18]: dataset_summary_df = pd.DataFrame(
    [
        {
            "dataset": "balanced training pool",
            "period": train_pool_df["period"].iloc[0],
            "rows": len(train_pool_df),
            "burned": int(train_pool_df["target"].sum()),
            "unburned": int((train_pool_df["target"] == 0).sum()),
            "burned_share": f"{train_pool_df['target'].mean():.1%}",
        },
        {
            "dataset": "↳ model training split",
            "period": train_model_df["period"].iloc[0],
            "rows": len(train_model_df),
            "burned": int(train_model_df["target"].sum()),
            "unburned": int((train_model_df["target"] == 0).sum()),
            "burned_share": f"{train_model_df['target'].mean():.1%}",
        },
        {
            "dataset": "↳ validation split",
            "period": validation_df["period"].iloc[0],
            "rows": len(validation_df),
            "burned": int(validation_df["target"].sum()),
            "unburned": int((validation_df["target"] == 0).sum()),
            "burned_share": f"{validation_df['target'].mean():.1%}",
        },
        {
            "dataset": "hold-out test (full landscape)",
            "period": test_df["period"].iloc[0],
            "rows": len(test_df),
            "burned": int(test_df["target"].sum()),
            "unburned": int((test_df["target"] == 0).sum()),
            "burned_share": f"{test_df['target'].mean():.1%}",
        },
    ]
)
```

```
display(Markdown("### Dataset summary"))
display(dataset_summary_df)
```

Dataset summary

	dataset	period	rows	burned	unburned	burned_share
0	balanced training pool	1997–2015	71824	35912	35912	50.0%
1	└ model training split	1997–2015	57459	28730	28729	50.0%
2	└ validation split	1997–2015	14365	7182	7183	50.0%
3	hold-out test (full landscape)	2016–2022	398697	5806	392891	1.5%

Encode features for modelling

Apply one-hot encoding to vegetation and select the final model columns. The encoding vocabulary is defined on the training split and reused everywhere to guarantee consistent column alignment.

```
In [19]: X_train = make_features(train_model_df)
FEATURE_COLUMNS = X_train.columns

X_valid = make_features(validation_df, FEATURE_COLUMNS)
X_test = make_features(test_df, FEATURE_COLUMNS)

y_train = train_model_df["target"]
y_valid = validation_df["target"]
y_test = test_df["target"]

display(Markdown(f"### Encoded feature matrix: {len(FEATURE_COLUMNS)} columns"))
display(list(FEATURE_COLUMNS))
```

Encoded feature matrix: 18 columns

```
['elevation',  
 'slope',  
 'aspect_eastness',  
 'aspect_northness',  
 'urban_distance',  
 'roads_distance',  
 'vegetation_3112',  
 'vegetation_3113',  
 'vegetation_313',  
 'vegetation_3114',  
 'vegetation_3115',  
 'vegetation_3116',  
 'vegetation_3111',  
 'vegetation_312',  
 'vegetation_3117',  
 'vegetation_322',  
 'vegetation_323',  
 'vegetation_324']
```

7. Model tuning

Learning objective: understand how grid search explores hyperparameter combinations and why we use a predefined validation fold.

What are hyperparameters?

A Random Forest has settings that *you* choose before training — these are called **hyperparameters**. They control how the forest is built (e.g., how many trees to grow, how deep each tree can be). The model cannot learn these from the data on its own; you must decide them. Different choices lead to models with different accuracy, so we need a principled way to find good values.

What is grid search?

Grid search is the simplest tuning strategy: we define a set of candidate values for each hyperparameter, and the algorithm tries *every* combination. For each combination it trains a model and measures performance on a validation set. The combination with the best score wins. It is exhaustive (guaranteed to find the best option in the grid) but can be slow when the grid is large.

In scikit-learn this is implemented by `GridSearchCV`.

Why a predefined validation fold instead of cross-validation?

Normally `GridSearchCV` would use **k-fold cross-validation**: it splits the data into k chunks, trains on $k-1$ of them, and validates on the remaining one, rotating through all chunks. Instead, we use `PredefinedSplit` to tell scikit-learn: "always use *this exact* validation set that we already separated earlier." This means:

1. Every hyperparameter combination is scored on the *same* held-out data, making results directly comparable.
2. It reuses the fixed validation subset we already carved out (Section 5), so tuning results are consistent with the validation scores we inspect elsewhere. Random k-fold would create new, different splits each time. Our data is **spatially autocorrelated** — nearby pixels share similar terrain, vegetation, and fire history. We are not considering spatial leakage here; this is an approximation for a first baseline.

Hyperparameter ranges explained

Parameter	Values	What it controls	Why these values
<code>n_estimators</code>	150, 300	Number of trees in the forest. More trees average out individual errors (reduce <i>variance</i>), but add computation time.	Beyond ~300 trees the improvement plateaus for a dataset of this size.
<code>max_features</code>	"sqrt", 0.5	How many features each tree considers at each split. Fewer features → more diverse trees → better ensemble.	"sqrt" ($\approx \sqrt{p}$ features) is the classic RF default; 0.5 (half the features) is a common alternative.
<code>min_samples_leaf</code>	1, 3, 5	Minimum number of samples required in a leaf node. Larger values prevent the tree from memorising rare patterns (acts as regularization).	1 = no constraint (fully grown); 3 and 5 progressively smooth the predictions.
<code>max_depth</code>	None, 15	Maximum number of splits from root to any leaf. Limits tree complexity.	None = grow until pure or <code>min_samples_leaf</code> is reached; 15 = moderate depth cap.

Scoring metric: ROC-AUC

We score each configuration with **ROC-AUC** (Area Under the Receiver Operating Characteristic curve). In plain terms, it answers: "If I pick one truly-burned pixel and one truly-unburned pixel at random, what is the probability that the model assigns a higher fire score to the burned one?" A value of 1.0 means perfect ranking; 0.5 means no better than chance.

ROC-AUC is useful here because it evaluates how well the model *ranks* pixels by fire susceptibility without requiring us to pick a specific classification threshold (we deal with thresholds later in Section 10).

```
In [20]: from sklearn.model_selection import GridSearchCV # exhaustive search over
from sklearn.model_selection import PredefinedSplit # use a fixed validation
```

```

from sklearn.ensemble import RandomForestClassifier # ensemble of decision

param_grid = {
    "n_estimators": [150, 300],
    "max_features": ["sqrt", 0.5],
    "min_samples_leaf": [1, 3, 5],
    "max_depth": [None, 15],
}

X_tuning = pd.concat([X_train, X_valid], ignore_index=True)
y_tuning = pd.concat([y_train, y_valid], ignore_index=True)

# PredefinedSplit: -1 = training row, 0 = validation row.
fold_labels = np.array([-1] * len(X_train) + [0] * len(X_valid))
validation_fold = PredefinedSplit(test_fold=fold_labels)

tuning_search = GridSearchCV(
    estimator=RandomForestClassifier(random_state=RANDOM_STATE, n_jobs=-1),
    param_grid=param_grid,
    scoring=["roc_auc"],
    refit="roc_auc",
    cv=validation_fold,
    n_jobs=1,
)
tuning_search.fit(X_tuning, y_tuning)

tuning_results_df = (
    pd.DataFrame(tuning_search.cv_results_)
    .sort_values("rank_test_roc_auc")
    [
        [
            "rank_test_roc_auc",
            "mean_test_roc_auc",
            "param_n_estimators",
            "param_max_features",
            "param_min_samples_leaf",
            "param_max_depth",
        ]
    ]
    .reset_index(drop=True)
)

display(Markdown("### Top tuning results (by validation ROC-AUC)"))
display(tuning_results_df.head(10))

```

Top tuning results (by validation ROC-AUC)

	rank_test_roc_auc	mean_test_roc_auc	param_n_estimators	param_max_features	p
0	1	0.842226	300	sqrt	
1	2	0.841920	300	0.5	
2	3	0.841856	300	sqrt	
3	4	0.841308	300	0.5	
4	5	0.841182	150	sqrt	
5	6	0.840813	150	sqrt	
6	7	0.840597	150	0.5	
7	8	0.840359	300	0.5	
8	9	0.840238	150	0.5	
9	10	0.840029	300	sqrt	

```
In [21]: best_params = tuning_search.best_params_
display(Markdown("### Selected hyperparameters"))
display(best_params)
```

Selected hyperparameters

```
{'max_depth': None,
 'max_features': 'sqrt',
 'min_samples_leaf': 1,
 'n_estimators': 300}
```

8. Model fitting

Learning objective: understand why we fit two models and which one is used for which purpose.

After tuning, we fit two models with the selected hyperparameters:

Model	Training data	Purpose
validation_model	80% training split only	Honest validation metrics (never saw validation rows)
rf_model	Full balanced pre-2016 pool	Final hold-out evaluation and mapping

The `GridSearchCV` object also refits internally, but that refit has seen the validation rows — so we do not use it for validation reporting.

```
In [22]: validation_model = RandomForestClassifier(
    **best_params,
    random_state=RANDOM_STATE,
    n_jobs=-1,
```

```

)
validation_model.fit(X_train, y_train)

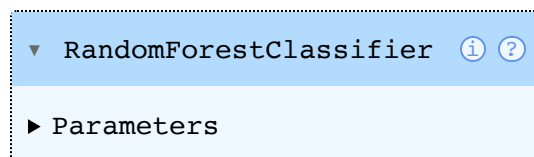
X_train_pool = make_features(train_pool_df, FEATURE_COLUMNS)
y_train_pool = train_pool_df["target"]

rf_model = RandomForestClassifier(
    **best_params,
    random_state=RANDOM_STATE,
    n_jobs=-1,
)
rf_model.fit(X_train_pool, y_train_pool)

display(Markdown("### Fitted Random Forest"))
display(rf_model)

```

Fitted Random Forest



A single tree from the forest

A Random Forest is an ensemble of many individual decision trees. To build intuition for what the model has learned, we visualise one tree extracted from the fitted forest.

Each **node** tests a single feature threshold:

- The left branch is taken when the condition is **True** ($\leq$ threshold).
- The right branch is taken when the condition is **False** ($>$ threshold).

Each **leaf** shows the majority class and the sample counts that ended up there during training. The leaf colour reflects the dominant class: orange-tinted = mostly burned, blue-tinted = mostly unburned.

A single tree can be quite deep and complex, so we cap the display at `max_depth=3` (the first three levels). This is enough to see the logic without an unreadable diagram. Vegetation one-hot columns appear as "vegetation_XXXX"; continuous predictors appear by name.

⚠ **One tree $\neq$ the ensemble.** This tree was built on a bootstrap sample of the training data and uses a random subset of features at each split. The ensemble averages over hundreds of such trees — each slightly different — which is what gives the Random Forest its robustness.

```

In [23]: from sklearn.tree import plot_tree # render a single decision tree as a dia
fig, ax = plt.subplots(figsize=(20, 7), constrained_layout=True)

```

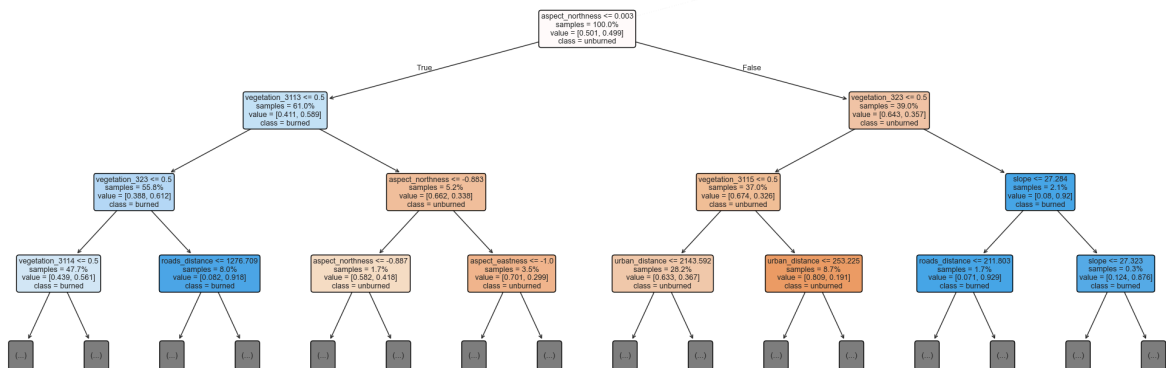
```

plot_tree(
    rf_model.estimators_[0],
    feature_names=list(FEATURE_COLUMNS),
    class_names=["unburned", "burned"],
    max_depth=3,
    filled=True,
    impurity=False,
    proportion=True,
    rounded=True,
    fontsize=9,
    ax=ax,
)
ax.set_title(
    "Single decision tree (tree #0, depth capped at 3) – one member of the F
    loc="left",
    fontweight="bold",
)

```

Out[23]: Text(0.0, 1.0, 'Single decision tree (tree #0, depth capped at 3) – one member of the Random Forest')

Single decision tree (tree #0, depth capped at 3) — one member of the Random Forest



9. Hold-out evaluation

Learning objective: understand why ranking metrics (ROC-AUC, recall@top%) are more appropriate than accuracy for susceptibility mapping, and why the hold-out set must keep real imbalance.

The key evaluation question:

Do 2016–2022 burned pixels receive higher susceptibility scores than unburned pixels on the same landscape?

We report:

- **ROC-AUC:** probability that a randomly chosen burned pixel scores higher than a randomly chosen unburned pixel (threshold-free).
- **Recall@Top K%:** if we prioritize the top K% of the landscape by score, what fraction of future burned pixels would that zone contain?

⚠️ **RF predict_proba is not a calibrated probability.** Random Forest vote fractions are biased toward 0.5 and reflect the balanced training prevalence, not the true fire probability. Use these scores strictly as a *ranking*, not as an estimate of $P(\text{fire})$.

```
In [24]: from sklearn.metrics import roc_auc_score      # area under the ROC curve
from sklearn.metrics import recall_score             # fraction of positives
from sklearn.metrics import RocCurveDisplay         # plot the ROC curve from
from sklearn.metrics import ConfusionMatrixDisplay  # plot the confusion matrix

def fire_score(model, X):
    """Return the model's susceptibility score (probability of class 1).

    Random Forest predict_proba returns [P(class=0), P(class=1)].
    We verify the expected class ordering as a safety check.
    """
    assert list(model.classes_) == [0, 1], (
        f"Expected classes [0, 1], got {list(model.classes_)}"
    )
    return model.predict_proba(X)[:, 1]

# Score the validation set (with the model that never saw validation rows).
validation_scores = validation_df[["target", "target_name", "row", "col", "x", "y"]].copy()
validation_scores["score"] = fire_score(validation_model, X_valid)

# Score the hold-out set (with the full pre-2016 model).
test_scores = test_df[["target", "target_name", "row", "col", "x", "y"]].copy()
test_scores["score"] = fire_score(rf_model, X_test)

score_summary_df = pd.DataFrame(
    [
        {
            "dataset": "validation (balanced pre-2016)",
            "rows": len(validation_scores),
            "burned_share": f"{validation_scores['target'].mean():.1%}",
            "mean_score": round(validation_scores["score"].mean(), 3),
        },
        {
            "dataset": "hold-out (full 2016–2022 landscape)",
            "rows": len(test_scores),
            "burned_share": f"{test_scores['target'].mean():.1%}",
            "mean_score": round(test_scores["score"].mean(), 3),
        },
    ]
)
display(Markdown("### Score summary"))
display(score_summary_df)
```

Score summary

	dataset	rows	burned_share	mean_score
0	validation (balanced pre-2016)	14365	50.0%	0.503
1	hold-out (full 2016–2022 landscape)	398697	1.5%	0.362

Metrics

```

In [25]: TOP_RECALL_SHARES = [0.10, 0.25, 0.50]

def recall_at_top_percent(y_true, score, top_share):
    """Share of burned pixels captured inside the highest-scoring map fraction

    This answers the operational question: if only this fraction of territory
    can be prioritized, how many observed fires would it have included?

    Ties at the cutoff make the mapped share slightly larger than requested,
    which is preferable to silently dropping equally scored pixels.
    """
    y_true = np.asarray(y_true)
    score = np.asarray(score)

    if not 0 < top_share <= 1:
        raise ValueError("top_share must be in the interval (0, 1].")

    threshold = np.quantile(score, 1 - top_share)
    return recall_score(y_true, score >= threshold, zero_division=0)

def compute_validation_metrics(name, y, score):
    """Metrics for the balanced validation sample (used for tuning only)."""
    return {
        "model": name,
        "evaluation": "balanced pre-2016 validation",
        "rows": len(y),
        "burned_share": f"{float(np.mean(y)):.1%}",
        "ROC-AUC": round(roc_auc_score(y, score), 3),
    }

def compute_holdout_metrics(name, y, score):
    """Metrics for the full-landscape hold-out (main performance summary)."""
    metrics = {
        "model": name,
        "evaluation": "full 2016-2022 landscape",
        "rows": len(y),
        "burned_share": f"{float(np.mean(y)):.1%}",
        "ROC-AUC": round(roc_auc_score(y, score), 3),
    }
    for top_share in TOP_RECALL_SHARES:
        key = f"Recall@Top{int(top_share * 100)}%"
        metrics[key] = round(recall_at_top_percent(y, score, top_share), 3)
    return metrics

In [26]: validation_metrics_df = pd.DataFrame(
    [compute_validation_metrics("tuned random forest", y_valid, validation_s
    ).set_index("model")
    display(Markdown("### Validation metrics (for tuning reference only)"))
    display(validation_metrics_df)

```

Validation metrics (for tuning reference only)

	evaluation	rows	burned_share	ROC-AUC
model				
tuned random forest	balanced pre-2016 validation	14365	50.0%	0.842

```
In [27]: holdout_metrics_df = pd.DataFrame(
    [compute_holdout_metrics("tuned random forest", y_test, test_scores["sco
    ]).set_index("model")
    display(Markdown("### Hold-out metrics (main performance summary)"))
    display(holdout_metrics_df)
```

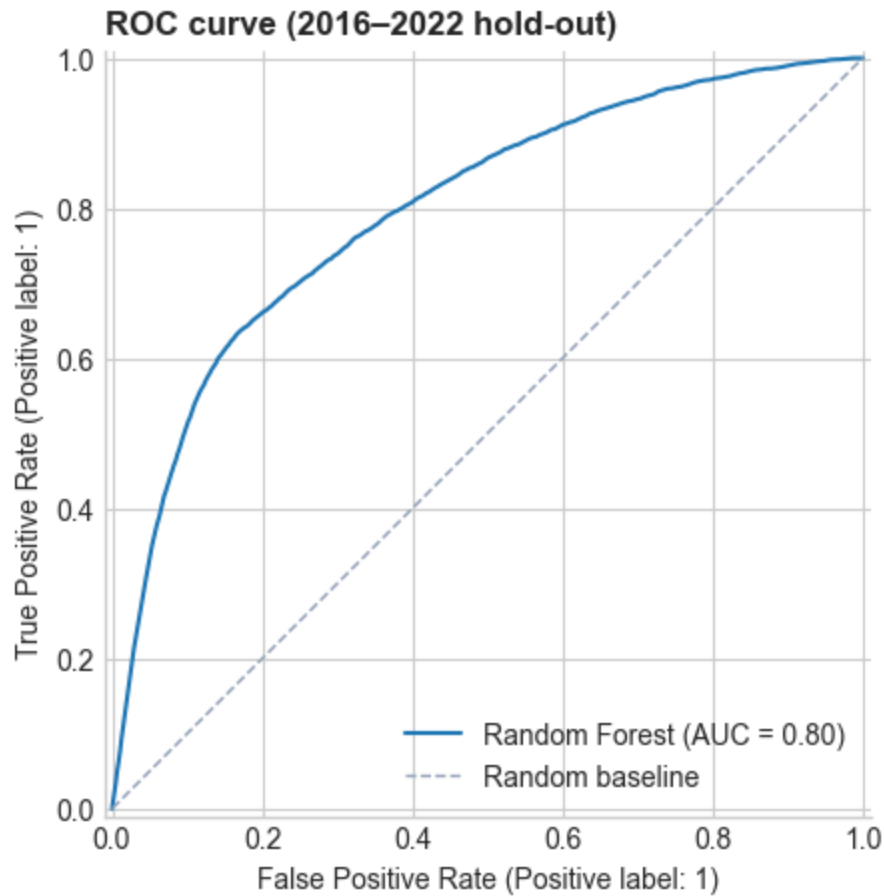
Hold-out metrics (main performance summary)

	evaluation	rows	burned_share	ROC-AUC	Recall@Top10%	Recall@Top25%	Re
model							
tuned random forest	full 2016–2022 landscape	398697	1.5%	0.798	0.495	0.698	

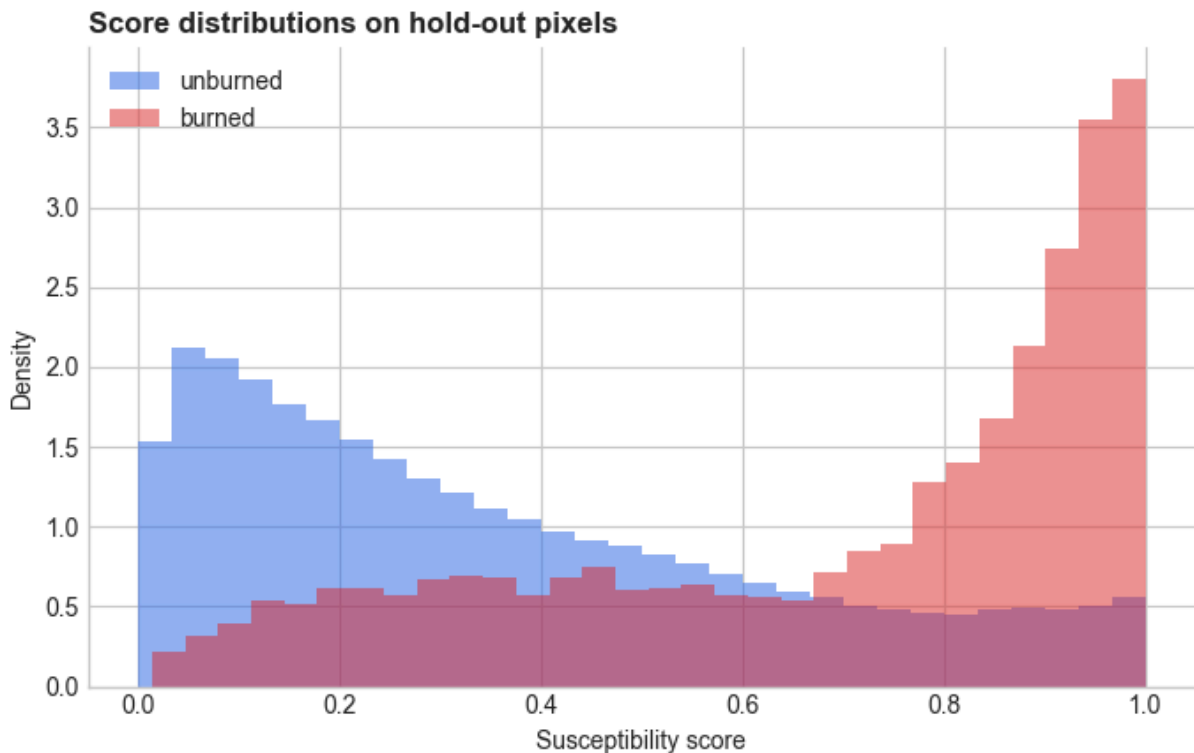
Reading the hold-out metrics. On the bundled data, the hold-out ROC-AUC is ~0.80: a randomly chosen burned pixel scores higher than a randomly chosen unburned pixel about 80% of the time. Recall@Top25% ~0.70 means that flagging the top quarter of the landscape by model score would have captured ~70% of 2016–2022 fires.

This is a useful ranking signal, but not a perfect predictor — which is expected when weather, ignition, and fuel moisture are absent from the model.

```
In [28]: fig, ax = plt.subplots(figsize=(6, 4.5), constrained_layout=True)
RocCurveDisplay.from_estimator(rf_model, X_test, y_test, name="Random Forest")
ax.plot([0, 1], [0, 1], linestyle="--", color="#94a3b8", linewidth=1, label="")
ax.set_title("ROC curve (2016–2022 hold-out)", loc="left", fontweight="bold")
ax.legend(loc="lower right", frameon=False)
clean_axes(ax)
```



```
In [29]: fig, ax = plt.subplots(figsize=(7, 4.5))
for target, label, color in [(0, "unburned", "#2563eb"), (1, "burned", "#dc2f2f")]:
    ax.hist(
        test_scores.loc[test_scores["target"] == target, "score"],
        bins=30,
        density=True,
        alpha=0.5,
        color=color,
        label=label,
    )
ax.set_title("Score distributions on hold-out pixels", loc="left", fontweight="bold")
ax.set_xlabel("Susceptibility score")
ax.set_ylabel("Density")
clean_axes(ax)
ax.legend(frameon=False)
fig.tight_layout()
```

10. Threshold decisions

Learning objective: understand that a threshold is a *decision rule*, not a property of the model, and that percentile-based thresholds are more portable than raw score cutoffs.

The Random Forest outputs a continuous score. Turning it into a binary class ("high susceptibility" vs "low") requires choosing a threshold. There is **no universal wildfire threshold**. The right choice depends on the cost of:

- Missing susceptible areas (false negatives → unmanaged risk)
- Over-flagging low-susceptibility areas (false positives → wasted resources)

Percentile thresholds

Rather than a raw score cutoff (which depends on model internals), define "high susceptibility" as a *territory share*: e.g., the top 10%, 25%, or 50% of valid pixels by model score. This is easier to explain to stakeholders and robust to model recalibration.

```
In [30]: baseline_burned_share = test_scores["target"].mean()

operational_threshold_rows = []
for high_risk_share in [0.10, 0.25, 0.50]:
    score_threshold = test_scores["score"].quantile(1 - high_risk_share)
    selected = test_scores["score"] >= score_threshold
    operational_threshold_rows.append(
        {
            "high_risk_territory_%": f"{high_risk_share:.0%}",
```

```

        "score_threshold": round(score_threshold, 3),
        "mapped_share": f"{float(selected.mean()):.1%}",
        "captured_burn_%": f"{recall_score(y_test, selected, zero_divisi
    )
    }
)

operational_threshold_df = pd.DataFrame(operational_threshold_rows)
display(Markdown("### Percentile thresholds on 2016–2022 hold-out"))
display(operational_threshold_df)

```

Percentile thresholds on 2016–2022 hold-out

	high_risk_territory_%	score_threshold	mapped_share	captured_burn_%
0	10%	0.810	10.0%	49.5%
1	25%	0.547	25.2%	69.8%
2	50%	0.293	50.3%	86.5%

The trade-off is clear: a smaller high-susceptibility zone is cheaper to inspect but misses more burned pixels. The top 10% captures ~50% of fires; the top 50% captures ~90%. The "right" threshold is a policy decision.

```

In [31]: medium_threshold = test_scores["score"].quantile(0.50)
high_threshold = test_scores["score"].quantile(0.75)

test_scores["susceptibility_class"] = pd.cut(
    test_scores["score"],
    bins=[-np.inf, medium_threshold, high_threshold, np.inf],
    labels=["low", "medium", "high"],
    include_lowest=True,
)

susceptibility_class_df = (
    test_scores.groupby("susceptibility_class", observed=True)
    .agg(
        pixels=("target", "size"),
        territory_share=("target", lambda v: len(v) / len(test_scores)),
        mean_score=("score", "mean"),
        burned_pixels=("target", "sum"),
        observed_burn_rate=("target", "mean"),
    )
    .reset_index()
)

susceptibility_class_df["captured_burn_share"] = (
    susceptibility_class_df["burned_pixels"] / test_scores["target"].sum()
)

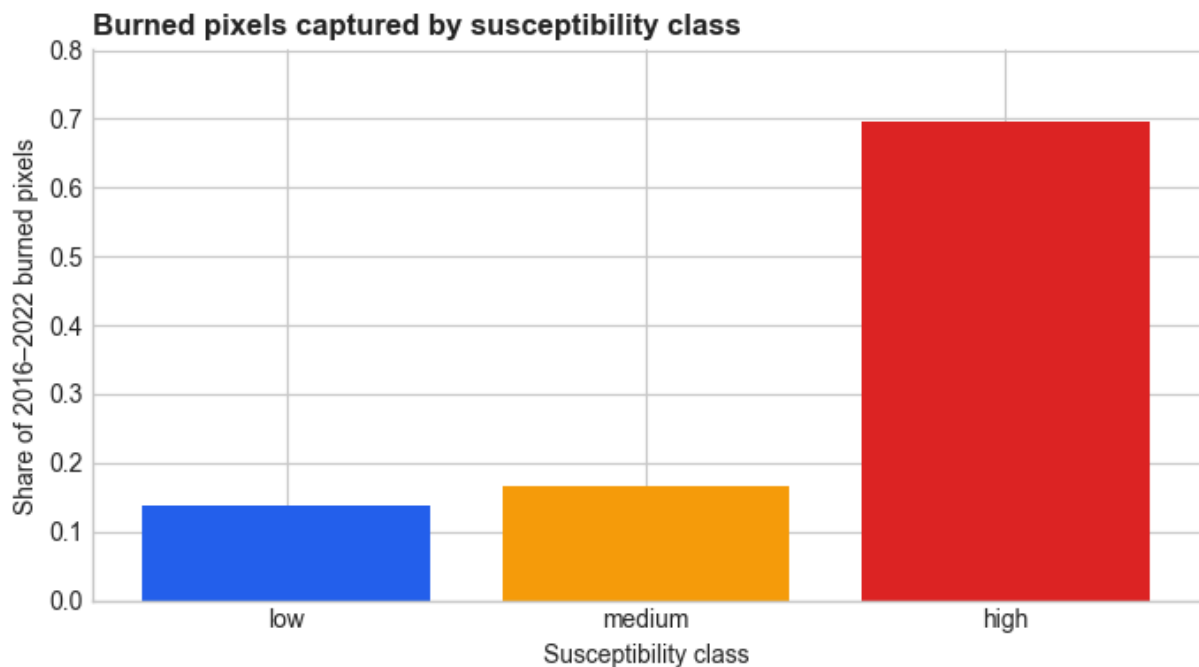
susceptibility_class_df["lift"] = (
    susceptibility_class_df["observed_burn_rate"] / baseline_burned_share
).round(1)
display(Markdown("### Susceptibility classes on hold-out"))
display(susceptibility_class_df)

```

Susceptibility classes on hold-out

	susceptibility_class	pixels	territory_share	mean_score	burned_pixels	observed_1
0	low	200005	0.501647	0.138124	806	
1	medium	99232	0.248891	0.411741	960	
2	high	99460	0.249463	0.763700	4040	

```
In [32]: fig, ax = plt.subplots(figsize=(7, 4))
class_plot_df = susceptibility_class_df.set_index("susceptibility_class").loc[
    ["low", "medium", "high"]
]
ax.bar(
    class_plot_df.index,
    class_plot_df["captured_burn_share"],
    color=["#2563eb", "#f59e0b", "#dc2626"],
)
ax.set_title("Burned pixels captured by susceptibility class", loc="left", f
ax.set_xlabel("Susceptibility class")
ax.set_ylabel("Share of 2016–2022 burned pixels")
ax.set_ylim(0, max(0.05, class_plot_df["captured_burn_share"].max() * 1.15))
clean_axes(ax)
fig.tight_layout()
```



Balanced confusion matrix (pedagogical diagnostic)

Confusion matrices are hard to interpret on the full landscape where unburned pixels outnumber burned pixels 20:1. For a clearer diagnostic, we subsample a 50/50 test set from the 2016–2022 hold-out.

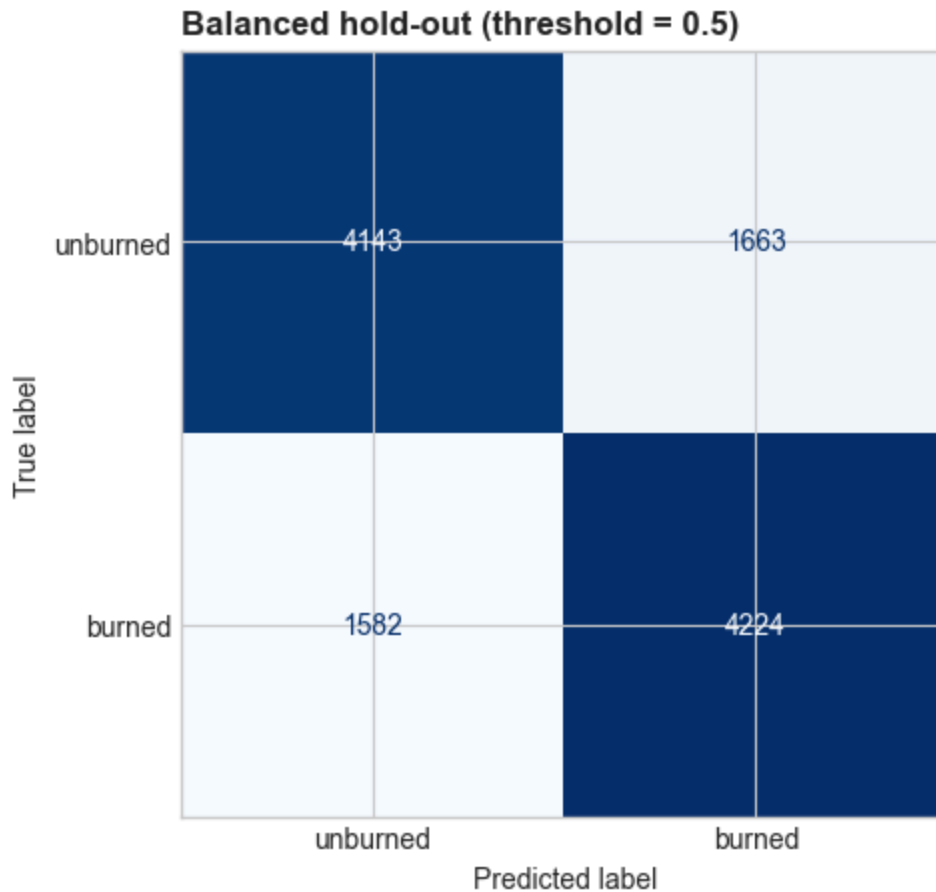
This balanced matrix shows whether the model has learned a real signal, but it does **not** represent landscape-level precision or false-alarm rates.

```
In [33]: burned_test_scores = test_scores.loc[test_scores["target"] == 1]
unburned_test_scores = test_scores.loc[test_scores["target"] == 0].sample(
    n=len(burned_test_scores),
    random_state=RANDOM_STATE,
)
balanced_test_scores = (
    pd.concat([burned_test_scores, unburned_test_scores], ignore_index=True)
    .sample(frac=1, random_state=RANDOM_STATE)
    .reset_index(drop=True)
)

balanced_confusion_threshold = 0.50
balanced_test_prediction = balanced_test_scores["score"] >= balanced_confusi

fig, ax = plt.subplots(figsize=(5, 4.5), constrained_layout=True)
ConfusionMatrixDisplay.from_predictions(
    balanced_test_scores["target"],
    balanced_test_prediction,
    display_labels=["unburned", "burned"],
    cmap="Blues",
    colorbar=False,
    ax=ax,
)
ax.set_title(
    f"Balanced hold-out (threshold = {balanced_confusion_threshold})",
    loc="left",
    fontweight="bold",
)
```

```
Out[33]: Text(0.0, 1.0, 'Balanced hold-out (threshold = 0.5)')
```



```
In [34]: tn, fp, fn, tp = (
    int((~balanced_test_prediction & (balanced_test_scores["target"] == 0)).s
    int((balanced_test_prediction & (balanced_test_scores["target"] == 0)).s
    int((~balanced_test_prediction & (balanced_test_scores["target"] == 1)).s
    int((balanced_test_prediction & (balanced_test_scores["target"] == 1)).s
)

balanced_metrics_df = pd.DataFrame(
    [
        {
            "threshold": balanced_confusion_threshold,
            "accuracy": round((tp + tn) / (tp + tn + fp + fn), 3),
            "sensitivity (recall)": round(tp / (tp + fn), 3),
            "specificity": round(tn / (tn + fp), 3),
            "precision": round(tp / (tp + fp), 3),
            "F1": round(2 * tp / (2 * tp + fp + fn), 3),
        }
    ]
)
display(Markdown("### Balanced confusion metrics"))
display(balanced_metrics_df)
```

Balanced confusion metrics

	threshold	accuracy	sensitivity (recall)	specificity	precision	F1
0	0.5	0.721	0.728	0.714	0.718	0.722

Both sensitivity and specificity around 70–75% confirm the model has learned a real discriminative signal. The balanced F1 ~ 0.72 is a decent baseline. However, for susceptibility mapping the full-landscape ROC-AUC and recall@top% metrics (Section 9) are the primary performance indicators.

11. Model explanation

Learning objective: understand permutation importance (MDA = Mean Decrease in Accuracy) as a model-specific explanation tool, and distinguish it from causal attribution.

How it works:

1. Shuffle one predictor column in the hold-out data.
2. Re-score the model.
3. Measure how much ROC-AUC drops compared to the unshuffled baseline.

A large drop means the model relied heavily on that variable for its ranking. We compute importance on the *hold-out set* (not training data) to measure what the model uses for generalization, not just for fitting.

⚠ Interpretation limits:

- Importance reflects model reliance, not ecological causality.
- Correlated predictors share importance (e.g., elevation and vegetation are confounded with geography).
- Vegetation class can proxy for fuel structure, land management, climate zone, and geography — permutation importance cannot separate these.

```
In [35]: from sklearn.inspection import permutation_importance # measure each feature

def raw_feature_auc(model, raw_X, y):
    """Custom scorer: encode raw features, then compute ROC-AUC."""
    encoded_X = make_features(raw_X, FEATURE_COLUMNS)
    return roc_auc_score(y, fire_score(model, encoded_X))

importance = permutation_importance(
    rf_model,
    test_df[FEATURES],
    y_test,
```

```

        scoring=raw_feature_auc,
        n_repeats=5,
        random_state=RANDOM_STATE,
        n_jobs=1,
    )

    importance_df = (
        pd.DataFrame(
            {
                "feature": FEATURES,
                "AUC_drop": importance.importances_mean.round(4),
                "std": importance.importances_std.round(4),
            }
        )
        .sort_values("AUC_drop", ascending=False)
        .reset_index(drop=True)
    )
    display(Markdown("### Permutation importance (hold-out ROC-AUC drop)"))
    display(importance_df)

```

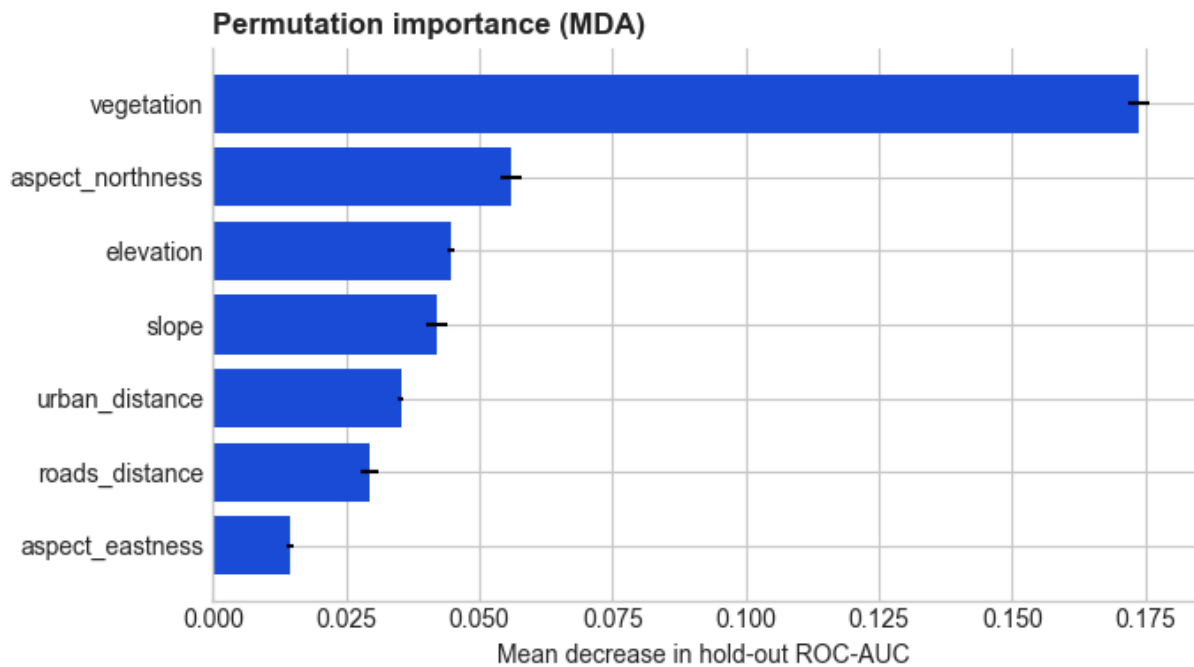
Permutation importance (hold-out ROC-AUC drop)

	feature	AUC_drop	std
0	vegetation	0.1737	0.0019
1	aspect_northness	0.0559	0.0020
2	elevation	0.0447	0.0008
3	slope	0.0420	0.0019
4	urban_distance	0.0353	0.0005
5	roads_distance	0.0295	0.0017
6	aspect_eastness	0.0144	0.0007

```

In [36]: fig, ax = plt.subplots(figsize=(7, 4))
        plot_df = importance_df.sort_values("AUC_drop")
        ax.barh(plot_df["feature"], plot_df["AUC_drop"], xerr=plot_df["std"], color=
        ax.set_title("Permutation importance (MDA)", loc="left", fontweight="bold")
        ax.set_xlabel("Mean decrease in hold-out ROC-AUC")
        clean_axes(ax)
        fig.tight_layout()

```



Vegetation is typically the dominant predictor, followed by aspect (northness), elevation, and slope. This is consistent with the exploratory analysis: vegetation class and south-facing exposure produce the clearest separation between burned and unburned samples in this Ligurian study area.

12. Susceptibility map

Learning objective: produce a spatial prediction and understand that the map is a model product, not independent evidence of accuracy.

The map uses `rf_model` (fitted on the full pre-2016 balanced pool with tuned hyperparameters). Red contours overlay the 2016–2022 burned areas for visual comparison, but the numeric metrics in Section 9 are the formal evaluation.

```
In [37]: grid_features = make_features(feature_table, FEATURE_COLUMNS)
grid_scores = fire_score(rf_model, grid_features)

susceptibility_grid = np.full(TEMPLATE_SHAPE, np.nan, dtype=float)
grid_rows = feature_table["row"].to_numpy()
grid_cols = feature_table["col"].to_numpy()
susceptibility_grid[grid_rows, grid_cols] = grid_scores

grid_score_summary = pd.Series(grid_scores).describe(percentiles=[0.1, 0.25,
display(Markdown("### Mapped susceptibility score summary"))
display(grid_score_summary)
```

Mapped susceptibility score summary


```

count      398697.000000
mean        0.362283
std         0.274410
min         0.000000
10%         0.056667
25%         0.130000
50%         0.293333
75%         0.546667
90%         0.810000
max         1.000000
dtype: float64

```

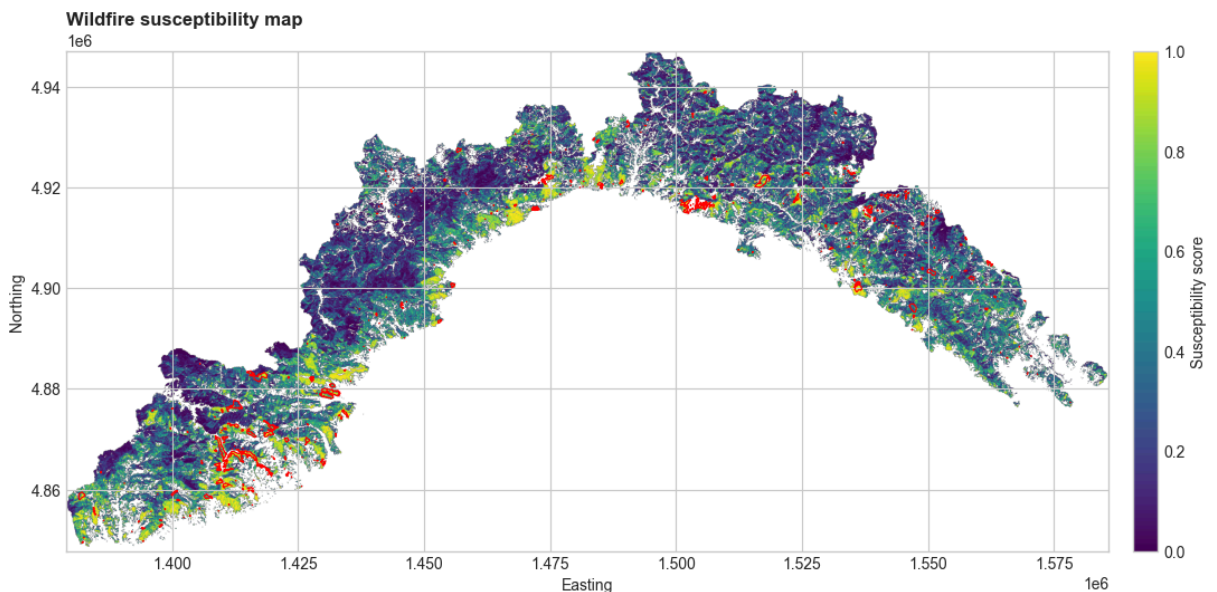
```

In [38]: fig, ax = plt.subplots(figsize=(12, 5.5))
         image = ax.imshow(
             susceptibility_grid, extent=extent, origin="upper", cmap="viridis", vmir
         )

         # Overlay: true 2016–2022 burned areas (visual comparison only).
         holdout_burned_overlay = test_burned_mask & valid_mask
         ax.contour(
             holdout_burned_overlay.astype(int),
             levels=[0.5],
             extent=extent,
             origin="upper",
             colors="red",
             linewidths=0.8,
         )

         ax.set_title("Wildfire susceptibility map", loc="left", fontweight="bold")
         ax.set_xlabel("Easting")
         ax.set_ylabel("Northing")
         ax.set_aspect("equal", adjustable="box")
         fig.colorbar(image, ax=ax, fraction=0.035, pad=0.02, label="Susceptibility s
         fig.tight_layout()

```



Take-home messages

1. **Raster → table:** Spatial raster data can be converted into a standard supervised-learning problem (one row per pixel, one column per feature).
2. **Temporal validation:** Training on past fires and evaluating on future fires is a stronger test than random splits, because it simulates the operational use case.
3. **Pseudo-absences:** Balanced sampling gives the model enough contrast to learn, but the resulting scores are rankings, not calibrated probabilities.
4. **Feature engineering matters:** Circular variables (aspect) need trigonometric decomposition; unordered categories (vegetation) need one-hot encoding.
5. **Evaluation metrics:** ROC-AUC and recall@top% are appropriate for ranking tasks. Accuracy is misleading under extreme class imbalance.
6. **Thresholds are decisions:** Converting a score into a class requires a policy choice. Percentile-based thresholds (top 10%, 25%, 50%) are more portable and interpretable than raw score cutoffs.
7. **Model explanation ≠ causality:** Permutation importance shows what the model relies on, not what drives fire occurrence.
8. **Spatial autocorrelation:** Nearby pixels share predictors and fire history. Random validation splits are optimistic; temporal and spatial hold-outs are more honest.
9. **The map is a model product:** It should be interpreted with the sampling design, predictor limitations, and validation results in mind.

Exercises

1. **Add climate predictors.** Extend `RASTER_PATHS`, `RASTER_FEATURES`, and `FEATURES` with: `temperature_summer.tif`, `temperature_winter.tif`, `precipitation_summer.tif`, `precipitation_winter.tif`. Re-run the workflow and compare hold-out ROC-AUC and recall@top%. Do the climate variables appear in the MDA table?
2. **Try another algorithm.** Replace `RandomForestClassifier` with `XGBClassifier`, `ExtraTreesClassifier`, or `LogisticRegression`. Keep the same splits. Compare validation metrics, hold-out metrics, and score distributions. Which model produces sharper score separation?
3. **Spatial cross-validation.** Design a geographic split (e.g., by easting or northing). Train on one part of the study area, test on the other. Compare with the temporal split. Does the model generalize spatially? What does a geographic split test that a temporal split does not?

4. **Vary the pseudo-absence ratio.** Try 1:2 and 1:5 unburned:burned ratios instead of 1:1. How does this change the score distribution, the ROC-AUC, and the confusion matrix? When might a higher ratio be preferable?